# Bangladesh Oil Gas and Mineral Corporation (BOGMC) – PetroBangla

## Post: Assistant Manager (CSE/IT/ICT)

**Exam Date:** 31.05.2024 | **Exam Taker:** BUET

# Written Exam — 2024

Complete Worked Solutions

These are worked solutions to the written questions from the Bangladesh Oil Gas and Mineral Corporation (BOGMC) – PetroBangla Assistant Manager (CSE/IT/ICT) exam, held on 31.05.2024 and conducted by BUET. Each question is reproduced first, followed by a clear, step-by-step solution and a highlighted final answer.

| Q# | Topic | Area |
|---|---|---|
| 1(a) | **Output of a Conditional Program** | Programming (C) |
| 1(b) | **Java Access Specifiers** | Object-Oriented Programming |
| 2(a) | **Binary Tree from Postorder** | Data Structures |
| 2(b) | **Quicksort — Sorted Input** | Algorithms |
| 3(a) | **Departments by Average Salary** | Database Management (SQL) |
| 3(b) | **Courses with More Than Two Offerings** | Database Management (SQL) |
| 4(a) | **Traffic Signal FSM — Flip-Flops** | Digital Logic Design |
| 4(b) | **Types of Software Testing** | Software Engineering |
| 5(a) | **Shell vs Kernel** | Operating Systems |
| 5(b) | **Handling RAID Drive Failure** | Storage Systems |
| 5(c) | **Use of mmWave in IoT** | Wireless / IoT |
| 5(d) | **Need for an Edge Server** | Cloud / IoT |

*How to use this guide: attempt each question yourself first, then compare your working against the solution shown.*

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**1(a)** *Programming  Conditionals*

**A C program is given. Write the output of the program when the value of a = 85.**

```
a = 85
if (a >= 90) print A;
if (a >= 80) print B;
if (a >= 70) print C;
if (a >= 60) print D; else print F;
```

**Solution**

Each **if** is a separate, independent test, and the **else** belongs only to the last one. For **a = 85**:

| Condition | Result | Output |
|---|---|---|
| a >= 90 | False | – |
| a >= 80 | True | **B** |
| a >= 70 | True | **C** |
| a >= 60 | True | **D** |

Because a >= 60 is true, the **else** branch (print F) is not executed.

**Answer: B C D (printed as BCD).**

**1(b)** *Object-Oriented Programming  Java*

**Java has three access specifiers: public, private and protected. Find which access specifier works with each of the Public, Private and Protected modes. Write Y if it does and N if it does not.**

**Solution**

| Access specifier | Public mode | Private mode | Protected mode |
|---|---|---|---|
| **public** | **Y** | N | N |
| **private** | N | **Y** | N |
| **protected** | N | N | **Y** |

*Note: in Java, public, private and protected are themselves the access modifiers/specifiers, so each one works only with its own mode.*

**Answer: public → Y for Public only; private → Y for Private only; protected → Y for Protected only.**

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**2(a)** *Data Structures  Binary Tree*

**You are given a binary tree with nodes a, b, c, d, e, f, g, h, i. The postorder of the binary tree is: a b f c h d e g i. Draw the binary tree and show the array representation of this binary tree (approximate).**

### Solution

Given nodes: **a, b, c, d, e, f, g, h, i**
Given postorder: **a b f c h d e g i**

In a postorder traversal the root is visited last, so **i** is the root. However, the postorder sequence alone does not say which nodes belong to the left subtree and which to the right.

> **Important:** a binary tree **cannot be uniquely constructed from its postorder traversal alone**. The **inorder traversal** (or equivalent additional information) is also required. So, for an exact tree and its array representation, the inorder sequence must be given.

**Answer: postorder alone gives insufficient information to determine the binary tree uniquely; the inorder sequence is also needed.**

**2(b)** *Algorithms  Quicksort*

**In the quicksort algorithm, taking the first element as the pivot, analyze the time complexity when all elements of the array are already sorted.**

### Solution

If the **first element is always chosen as the pivot** and the array is **already sorted**, the pivot is always the smallest (or largest) element. Each partition puts all remaining elements on one side, so the partitions are highly unbalanced.

$$T(n) = T(n-1) + O(n)$$

$$T(n) = n + (n-1) + \ldots + 2 + 1 = \frac{n(n+1)}{2} = O(n^2)$$

**Answer: worst-case time complexity = $O(n^2)$.**

**3** *Database Management  SQL*

**Consider the following relational schema (PK = primary key, FK = foreign key). Write SQL queries for parts (a) and (b).**

```
Employee   (empno PK, empname, monthlysalary, deptno FK, mgrno FK)
Department (deptno PK, deptname, deptlocation)
Course     (crscode PK, crsdesc, crscategory, crsduration)
Offering   (offbegindate, crscode FK, offeringlocation, empno FK)
```

*Attribute names are written in a consistent form; the keys are as given in the question.*

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**3(a)** *Database Management  GROUP BY / HAVING*

**Find the departments whose average monthly salary is greater than 1000.**

**Solution**

```
SELECT   deptno, AVG(monthlysalary) AS avg_salary
FROM     Employee
GROUP BY deptno
HAVING   AVG(monthlysalary) > 1000;
```

**Answer: the query above lists each department with its average salary, keeping only those above 1000.**

**3(b)** *Database Management  GROUP BY / HAVING*

**Find the courses that have more than two offerings.**

**Solution**

```
SELECT   crscode, COUNT(*) AS offering_count
FROM     Offering
GROUP BY crscode
HAVING   COUNT(*) > 2;
```

**Exam point:** use **WHERE** for row-level conditions and **HAVING** for conditions on grouped/aggregate results.

**Answer: the query above returns each course code offered more than twice, with its offering count.**

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**4(a)** *Digital Logic Design  Finite State Machine*

**A traffic signal cycles from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle, GREEN is on for 70 seconds, YELLOW for 5 seconds and RED for 75 seconds. The traffic light is implemented as a finite state machine (FSM) whose only input is a clock of 5-second period. What is the minimum number of flip-flops required to implement this FSM?**

### Solution

Clock period = **5 seconds**, so each state lasts a whole number of clock cycles.

| Light | On time | Clock cycles (time ÷ 5 s) |
|---|---|---|
| GREEN | 70 s | 70 / 5 = 14 |
| YELLOW | 5 s | 5 / 5 = 1 |
| RED | 75 s | 75 / 5 = 15 |
| **Total states** | | **14 + 1 + 15 = 30** |

For $n$ flip-flops the FSM can have up to $2^n$ states, so we need $2^n \geq 30$. Since $2^4 = 16$ is too small and $2^5 = 32 \geq 30$, $n = 5$.

**Answer: 5 flip-flops.**

**4(b)** *Software Engineering  Testing*

**Five software engineering scenarios are given for Unit testing, Regression testing, Smoke testing, Integration testing and Load testing (one is done for you). Write the name of the testing and whether it is functional, non-functional, or both.**

### Solution

| Testing | What it checks | Type |
|---|---|---|
| **Unit testing** | Each individual function, method or class in isolation. | **Functional** |
| **Integration testing** | That separately tested modules work correctly together and exchange data properly. | **Functional** |
| **Smoke testing** | A quick check that the most critical functions of a new build work, to decide if deeper testing is worthwhile. | **Functional** |
| **Regression testing** | That existing features still work after a code change, fix or new feature. | **Functional** |
| **Load testing** | System behaviour and response time under the expected (or heavy) number of users or requests. | **Non-functional** |

*Note: the five scenario descriptions are not part of the question text available here, so the table gives the standard classification of each testing type. Match each scenario to the type it describes.*

**Answer: Unit, Integration, Smoke and Regression testing are functional; Load testing is non-functional.**

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**5(a)** *Operating Systems  Shell and Kernel*

**What is the difference between a shell and a kernel?**

**Solution**

| Shell | Kernel |
|---|---|
| Interface between the **user and the kernel** | Core of the **operating system** |
| Accepts and interprets user commands | Manages CPU, memory, devices and processes |
| Runs as a user-level program | Runs with privileged/system-level access |
| Example: **Bash, Zsh** | Example: **Linux kernel** |

**Answer: the shell is the user-facing command interpreter; the kernel is the privileged core that manages the hardware and processes.**

**5(b)** *Storage Systems  RAID*

**How do you recover when a drive fails in a RAID array?**

**Solution**

It depends on the RAID level:

- **RAID 1:** replace the failed drive and **rebuild (re-mirror)** the data from the surviving drive.
- **RAID 5:** replace the failed drive; the data is **reconstructed using parity** and the array is rebuilt.
- **RAID 6:** can tolerate **two drive failures**; replace the failed drive(s) and rebuild using dual parity.
- **RAID 0:** **no fault tolerance**; a single drive failure causes data loss.

**General steps:** identify the failed drive → replace it → rebuild/reconstruct the RAID array → verify array health.

**Answer: replace the failed drive and rebuild from mirror or parity (RAID 1/5/6); RAID 0 cannot recover.**

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**5(c)** *Wireless / IoT  mmWave*

### What is the use of mmWaves in IoT?

**Solution**

**mmWave (millimetre wave)** communication provides **very high bandwidth and data rates** with low latency. In IoT it is used for:

- High-speed communication
- High-density IoT deployments
- **5G-based IoT**
- Radar and sensing applications
- Smart factories and autonomous systems

**Answer: mmWave gives IoT very high data rates and low latency for dense, real-time, 5G-based and sensing applications.**

**5(d)** *Cloud / IoT  Edge Computing*

### What is the need for an edge server?

**Solution**

An **edge server** processes data **near the source/device** instead of sending everything to a central cloud server.

**Benefits**

- Reduces **latency**
- Reduces **network bandwidth usage**
- Enables faster real-time processing
- Improves reliability when cloud connectivity is limited
- Can improve data privacy by processing data locally

**Example:** in a smart factory, an edge server can analyse sensor data locally and immediately detect a machine fault.

**Answer: edge servers cut latency and bandwidth use by processing data close to where it is produced.**

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